

LCA OF THE IMPOSSIBLE

HYDRA OF LERNA

WHEN REGENERATION BURNS

One wound. Two heads.

EPISODE #56



STEP ONE

A MONSTER. A CONTROLLED EVENT.

The model treats the Hydra not as magic, but as a hazardous biological incident: a multi-headed organism neutralised by cutting, cauterisation and controlled containment.



FUNCTIONAL UNIT

One Hydra neutralisation and containment event.

REFERENCE FLOW

One 3.2 t, nine-headed canonical Hydra.

TIMEFRAME

One field response cycle, from arrival to sealed head.

Geography	Lerna / UK EF	Proxy
Model type	Narrative LCA	Screening
Technology	Thermal cauterisation	Baseline
Result unit	kg CO ₂ e	GWP



The service is not killing a myth. It is preventing regeneration.

STEP ONE

BOUNDARY: FIRE BEFORE REGROWTH

INCLUDED

- ✓ Response travel
- ✓ Field equipment freight
- ✓ Propane cauterisation
- ✓ Diesel pump/generator
- ✓ Containment materials

EXCLUDED

- ✗ Hero metabolism
- ✗ Divine intervention
- ✗ Venom afterlife
- ✗ Long-term marsh remediation



CUTOFF EVENT

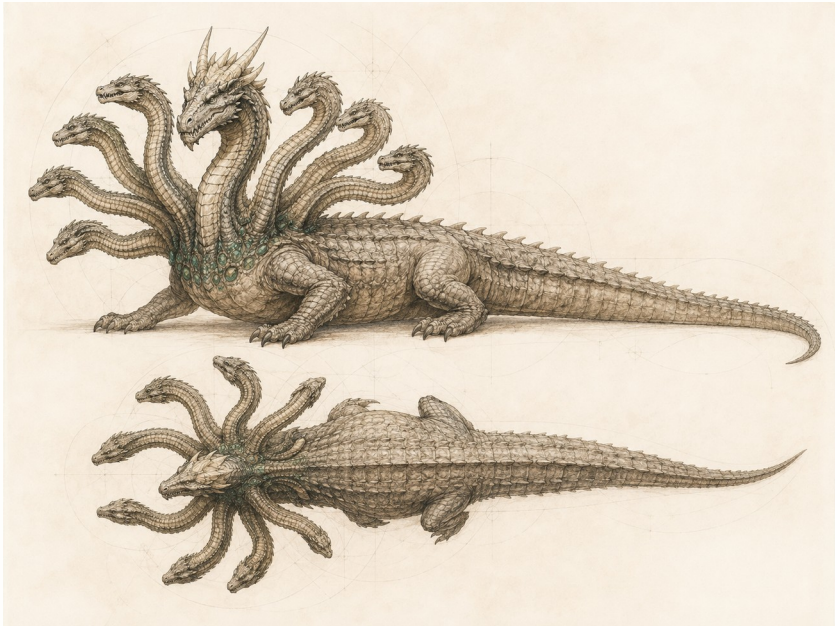
The model stops once the immortal head is sealed under the boulder and the field containment pit is closed.



The boundary follows the intervention, not the legend's afterlife.

STEP TWO

NINE HEADS. ONE BODY PLAN.



11 m reconstructed body length

Mass proxy	3.2 t
Heads	8 mortal + 1 immortal
Body type	Semi-aquatic reptilian
Main material	Living biomass excluded
Containment	Lime + LDPE + boulder

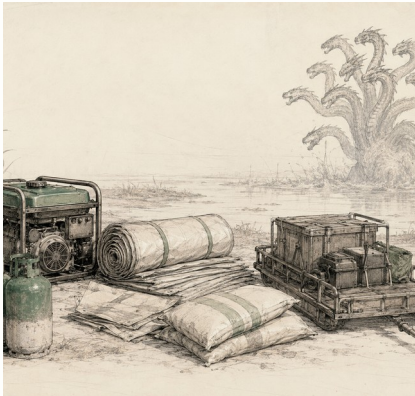


The geometry is frozen before the inventory is calculated.

STEP TWO

THE INVENTORY BEHIND THE LABOUR

FLOW	QTY	BASIS
Propane	190 L	cauterisation
Diesel	80 L	field power
Quicklime	250 kg	containment
LDPE sheet	35 kg	liner/wrap
Wood	80 kg	stakes/firebrands
Metal tools	12 kg	allocated
Freight	1,120 t·km	4 t x 280 km
Van	280 km	response travel



EXPECTED HOTSPOT

Fuel and field energy dominate before the myth becomes visible.



Every inventory line has quantity, unit, factor and basis.

STEP TWO

FROM CUT TO CONTAINMENT



01

Locate

Marsh access
and safe
perimeter.

02

Immobilise

Reduce movement
and expose neck
crown.

03

Sever

Remove mortal
heads one at a
time.

04

Cauterise

Seal stumps
before regrowth.

05

Contain

Lime, wrap and
bury the
immortal head.



The modelled process is a field-control chain, not a duel.

STEP TWO

FUEL MOVES THE STORY



PROPANE

190 L

DIESEL

80 L

FREIGHT

1,120 t·km

ELECTRICITY

60 kWh

TRANSPORT CONVERSION

 $4.0 \text{ t equipment and materials} \times 280 \text{ km} = 1,120 \text{ t·km.}$

A distance in kilometres becomes freight only after mass is declared.

STEP THREE

FACTORS BEFORE RESULTS

FLOW	FACTOR	UNIT	SOURCE
Propane combustion	1.54358	kg CO ₂ e/L	DEFRA 2026
Propane WTT	0.18170	kg CO ₂ e/L	DEFRA 2026
Diesel combustion	2.58354	kg CO ₂ e/L	DEFRA 2026
Diesel WTT	0.61101	kg CO ₂ e/L	DEFRA 2026
Quicklime proxy	1.09668	kg CO ₂ e/kg	Sustainability 2023
Average plastics	3.17051	kg CO ₂ e/kg	DEFRA 2026
Freight HGV	0.19947	kg CO ₂ e/t·km	DEFRA 2026
UK electricity	0.13096	kg CO ₂ e/kWh	DEFRA 2026

Proxy disclosure: quicklime is not in DEFRA 2026; a published lime-production proxy is used and tested in sensitivity B.



A short factor label links to a complete external register.

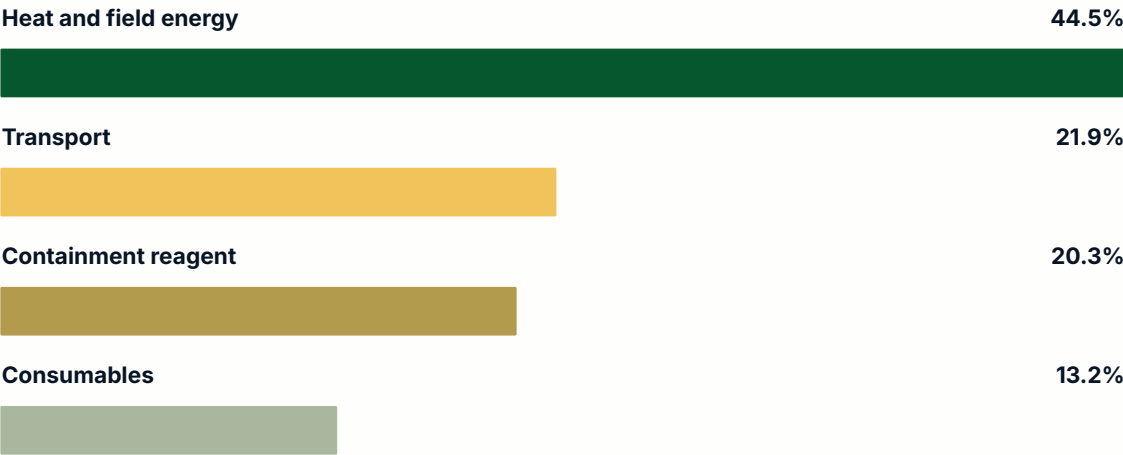
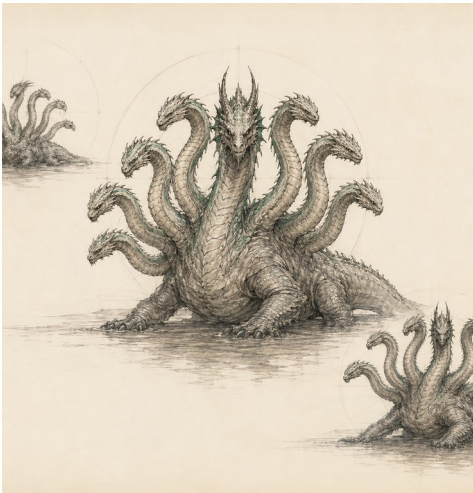
STEP FOUR

THE FOOTPRINT OF REGENERATION

BASELINE TOTAL

1.35 t CO₂e

per neutralised Hydra event



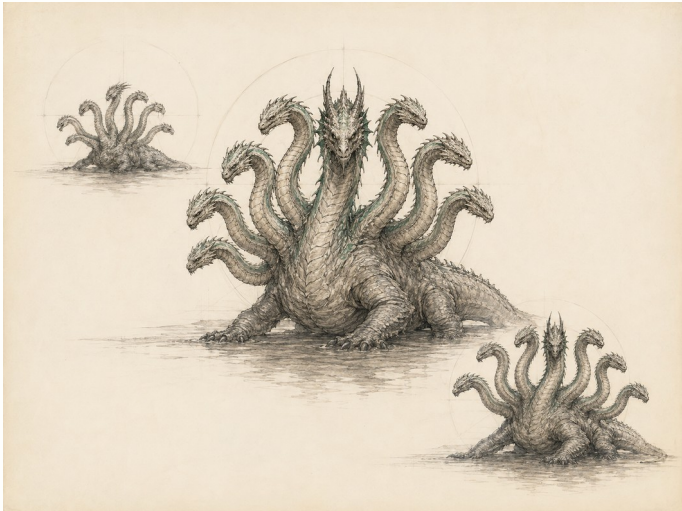
HOTSPOT: field heat and energy account for 44.5% of the baseline footprint.



The dominant burden is operational control, not the creature itself.

STEP FIVE

IF THE BEAST IS LARGER



Low mass

2.2 t body



1.01 t CO₂e

Baseline

3.2 t body



1.35 t CO₂e

High mass

4.8 t body



1.88 t CO₂e

High mass raises the total by +39.8% because fuel, lime, sheeting and freight scale together.



Changing mass changes the intervention system, not only the animal.

STEP FIVE

IF THE LIME FACTOR MOVES

LOW PROXY

1.26 t CO₂e

lime EF

0.750 kg CO₂e/kg**-6.4%**

BASELINE

1.35 t CO₂e

lime EF

1.09668 kg CO₂e/kg**0.0%**

HIGH PROXY

1.37 t CO₂e

lime EF

1.17696 kg CO₂e/kg**+1.5%**

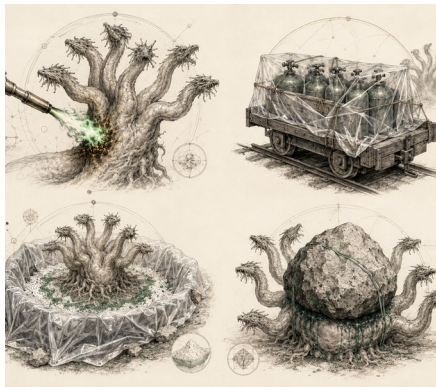
The quicklime proxy matters, but it does not overturn the result: energy and logistics remain dominant.



A proxy can move the answer without changing the story.

STEP SIX

FINDING. CONTROL HAS A FOOTPRINT

**01 Regrowth creates energy demand.**

Propane and field power drive 44.5% of the baseline.

02 Containment is material.

Quicklime and LDPE are 28.6% together.

03 Transport is visible.

Declared freight and van travel contribute 21.9%.

04 Scale changes everything.

A 4.8 t Hydra reaches 1.88 t CO₂e.

“Cutting the head is easy; sealing the system is costly.”

A self-replicating problem demands a second system to stop it.

SEASON II MYTHS & LEGENDS

Where impossible stories become measurable systems.

HYDRA OF LERNA